CONVEXITY AND SVMS PROBLEM SHEET

This sheet covers convexity, support vector machines and kernels. Questions 1 and 2 are about convex functions and Jensen's inequality. Question 3 solves a tiny SVM by hand, in both the primal and the dual form. Question 4 asks which functions are valid kernels.

1

(a) Starting from the definition of a convex function

$$f(ax + (1-a)y) \leq af(x) + (1-a)f(y) \tag{1}$$

Let $a = \epsilon/(x-y)$ and rearrange the inequality to give

$$(x-y)\left(\frac{f(y+\epsilon) - f(y)}{\epsilon}\right)$$

on the left-hand side. Taking the limit $\epsilon \to 0$ show that the function $f(x)$ lies above the tangent line $t(x) = f(y) + (x-y)f'(y)$ going through the point y. [4 marks]

Rearranging the Equation 1

$$f(y + a(x-y)) \leq f(y) + a(f(x) - f(y)).$$

Or

$$\frac{f(y + a(x-y)) - f(y)}{a} \leq f(x) - f(y).$$

Letting $a = \epsilon/(x-y)$ then

$$(x-y)\frac{f(y+\epsilon) - f(y)}{\epsilon} \leq f(x) - f(y)$$

Taking the limit $\epsilon \to 0$ then using

$$\lim_{\epsilon \to 0} \frac{f(y+\epsilon) - f(y)}{\epsilon} = f'(y)$$

So that

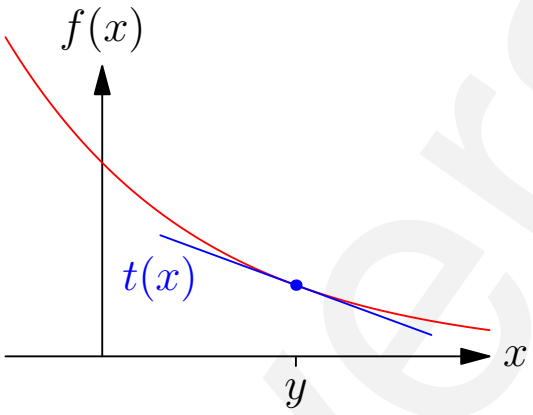
$$(x-y)f'(y) \leq f(x) - f(y)$$

or

$$f(x) \geq f(y) + (x-y)f'(y) = t(x).$$

Answers

(b) Sketch the tangent line, $t(x)$, at the point y in the graph shown below. [1 mark]



(c) Starting from the inequality for a convex function

$$f(x) \geq f(y) + (x - y)\, f'(y) \tag{2}$$

consider the case $y = x + \epsilon$, then by Taylor expanding $f(x + \epsilon)$ and $f'(x + \epsilon)$ around x and keeping all terms up to order ϵ^2 show that for a convex function $f''(x) \geq 0$. [4 marks]

We use the expansions

$$f(x + \epsilon) = f(x) + \epsilon f'(x) + \frac{\epsilon^2}{2} f''(x) + O(\epsilon^3)$$
$$f'(x + \epsilon) = f'(x) + \epsilon f''(x) + O(\epsilon^2)$$

Using $y = x + \epsilon$ and substituting into the Equation (2)

$$f(x) \geq f(x + \epsilon) - \epsilon f'(x + \epsilon)$$
$$f(x) \geq f(x) + \epsilon f'(x) + \frac{\epsilon^2}{2} f''(x) + O(\epsilon^3) - \epsilon\big(f'(x) + \epsilon f''(x) + O(\epsilon^2)\big)$$

Or subtracting $f(x)$ from both sides

$$0 \geq -\frac{\epsilon^2}{2} f''(x) + O(\epsilon^3)$$

Since this has to be true for all $\epsilon > 0$ this requires $f''(x) \geq 0$.

(d) Prove that $-\log(x)$ is convex-up for $x > 0$. [1 mark]

$$\frac{\mathrm{d}^2(-\log(x))}{\mathrm{d}x^2} = \frac{1}{x^2} \geq 0$$

End of question 1

Answers

2

(a) If $\|\boldsymbol{x}\|$ is a proper norm use the triangular inequality ($\|\boldsymbol{x}+\boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$), "linearity" of a norm ($\|a\boldsymbol{x}\| = |a|\|\boldsymbol{x}\|$) and the definition of convexity, to show that the norm is convex. [5 marks]

For any vectors, $\boldsymbol{x}$, and $\boldsymbol{y}$ and any scalar $a \in [0,1]$ then

$$\|a\boldsymbol{x} + (1-a)\boldsymbol{y}\| \leq \|a\boldsymbol{x}\| + \|(1-a)\boldsymbol{y}\| = a\|\boldsymbol{x}\| + (1-a)\|\boldsymbol{y}\|$$

where the first inequality follows from the triangular inequality and the second equality from the "linearity" of the norm (with $a, 1-a \geq 0$). However,

$$\|a\boldsymbol{x} + (1-a)\boldsymbol{y}\| \leq a\|\boldsymbol{x}\| + (1-a)\|\boldsymbol{y}\|$$

is the defining equation of convexity.

Answers

(b) Consider a classification problem where $\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})$ is the probability that a learning machine with parameters $\boldsymbol{\theta}$ predicts that input $\boldsymbol{x}$ belongs to class $c \in \mathcal{C}$. Assume the training is stochastic so the probability of obtaining parameters $\boldsymbol{\theta}$ is $\rho(\boldsymbol{\theta})$. Let $\hat{m}_c(\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{\theta}}\Big[\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})\Big]$ be the output of the mean machine for class c. Assuming that for a data point $(\boldsymbol{x},y)$, where y is a class label, we use a cross entropy loss

$$L(\boldsymbol{x},y,\boldsymbol{\theta}) = -\sum_{c\in\mathcal{C}} [\![y=c]\!] \log\Big(\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})\Big),$$

show that the expected loss over inputs and parameters can be written as the expected loss of the mean machine plus a second loss. Use Jensen's inequality ($\mathbb{E}\big[\log(X)\big] \leq \log\big(\mathbb{E}\big[X\big]\big)$) to show the second term is non-negative. [5 marks]

$$\bar{L} = \mathbb{E}_{(\boldsymbol{x},y)}\left[\mathbb{E}_{\boldsymbol{\theta}}\left[-\sum_{c\in\mathcal{C}} [\![y=c]\!] \log\Big(\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})\Big)\right]\right]$$

$$= -\mathbb{E}_{(\boldsymbol{x},y)}\left[\sum_{c\in\mathcal{C}} [\![y=c]\!] \log(\hat{m}_c(\boldsymbol{x}))\right] - \mathbb{E}_{(\boldsymbol{x},y)}\left[\mathbb{E}_{\boldsymbol{\theta}}\left[\sum_{c\in\mathcal{C}} [\![y=c]\!] \log\left(\frac{\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})}{\hat{m}_c(\boldsymbol{x})}\right)\right]\right]$$

The first term acts like a bias. The second (variance-like) term is

$$-\mathbb{E}_{(\boldsymbol{x},y)}\left[\sum_{c\in\mathcal{C}} [\![y=c]\!] \mathbb{E}_{\boldsymbol{\theta}}\Big[\log\Big(\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})\Big)\Big]\right] + \mathbb{E}_{(\boldsymbol{x},y)}\left[\sum_{c\in\mathcal{C}} [\![y=c]\!] \log(\hat{m}_c(\boldsymbol{x}))\right]$$

But using Jensen's inequality
$\mathbb{E}_{\boldsymbol{\theta}}\Big[\log\Big(\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})\Big)\Big] \leq \log\Big(\mathbb{E}_{\boldsymbol{\theta}}\Big[\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})\Big]\Big) = \log(\hat{m}_c(\boldsymbol{x}))$. Thus this second term is non-negative.

End of question 2

Answers

3 In the lectures the maximum-margin SVM finds $\boldsymbol{w}$ and b that minimise $\frac{1}{2}\|\boldsymbol{w}\|^2$ subject to $y_k(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_k - b) \geq 1$ for every data point. The dual problem is to maximise

$$\mathcal{L}(\boldsymbol{\alpha}) = \sum_k \alpha_k - \frac{1}{2}\sum_{k,l} \alpha_k \alpha_l y_k y_l \boldsymbol{x}_k^\mathsf{T}\boldsymbol{x}_l$$

subject to $\sum_k \alpha_k y_k = 0$ and $\alpha_k \geq 0$, after which $\boldsymbol{w} = \sum_k \alpha_k y_k \boldsymbol{x}_k$. In this question we solve a tiny problem by hand.

(a) Take two points, $\boldsymbol{x}_1 = (1,1)^\mathsf{T}$ with $y_1 = +1$ and $\boldsymbol{x}_2 = (-1,-1)^\mathsf{T}$ with $y_2 = -1$. By symmetry the solution has $b = 0$ and $\boldsymbol{w} = (u,u)^\mathsf{T}$. Find u and the margin $1/\|\boldsymbol{w}\|$. Check that the margin is half the distance between the two points.

Both constraints read $2u \geq 1$, and we want u as small as possible, so $u = \frac{1}{2}$ and $\boldsymbol{w} = (\frac{1}{2},\frac{1}{2})^\mathsf{T}$. The margin is $1/\|\boldsymbol{w}\| = 1/(1/\sqrt{2}) = \sqrt{2}$. The points are $\|\boldsymbol{x}_1 - \boldsymbol{x}_2\| = \|(2,2)^\mathsf{T}\| = 2\sqrt{2}$ apart, and half of that is $\sqrt{2}$.

[2 marks]

(b) Now solve the dual problem for the same two points. Show that the constraint forces $\alpha_1 = \alpha_2 = \alpha$, write $\mathcal{L}$ as a function of α and maximise it. Recover $\boldsymbol{w}$ and check that it agrees with part (a). Find b from the fact that $\boldsymbol{x}_1$ lies on the margin, $y_1(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_1 - b) = 1$.

$\alpha_1 y_1 + \alpha_2 y_2 = \alpha_1 - \alpha_2 = 0$, so $\alpha_1 = \alpha_2 = \alpha$. The inner products are $\boldsymbol{x}_1^\mathsf{T}\boldsymbol{x}_1 = \boldsymbol{x}_2^\mathsf{T}\boldsymbol{x}_2 = 2$ and $\boldsymbol{x}_1^\mathsf{T}\boldsymbol{x}_2 = -2$, so the double sum is

$$\alpha^2(2 + 2 + 2 \times (+1)(-1)(-2)) = 8\alpha^2, \qquad \mathcal{L}(\alpha) = 2\alpha - 4\alpha^2.$$

$\mathcal{L}'(\alpha) = 2 - 8\alpha = 0$ gives $\alpha = \frac{1}{4}$. Then $\boldsymbol{w} = \frac{1}{4}(1,1)^\mathsf{T} - \frac{1}{4}(-1,-1)^\mathsf{T} = (\frac{1}{2},\frac{1}{2})^\mathsf{T}$, as in part (a). Finally $\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_1 - b = 1 - b = 1$, so $b = 0$.

[3 marks]

(c) Add a third point, $\boldsymbol{x}_3 = (3,0)^\mathsf{T}$ with $y_3 = +1$. Without solving the problem again, explain why the solution does not change, and why $\alpha_3 = 0$.

With the old solution, $y_3(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_3 - b) = 1.5 > 1$, so the new constraint is already satisfied. Adding a constraint that the old optimum satisfies cannot give a better optimum, so the solution is unchanged. Complementary slackness, $\alpha_3\big(y_3(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_3 - b) - 1\big) = 0$ with the bracket equal to 0.5, forces $\alpha_3 = 0$. Only the support vectors (points on the margin) have $\alpha_k > 0$.

[2 marks]

(d) Classify the new point $\boldsymbol{x} = (-2,1)^\mathsf{T}$ using the dual form $\operatorname{sgn}\big(\sum_k \alpha_k y_k \boldsymbol{x}_k^\mathsf{T}\boldsymbol{x} - b\big)$.

The α_3 term vanishes, so the argument is $\frac{1}{4}(1,1)(-2,1)^\mathsf{T} - \frac{1}{4}(-1,-1)(-2,1)^\mathsf{T} = -\frac{1}{4} - \frac{1}{4} = -\frac{1}{2}$: class -1.

[1 mark]

Answers

(e) Return to the two points of part (a), but use a soft margin with $C = 0.1$, so that the dual constraint becomes $0 \le \alpha_k \le C$. Find α, $\boldsymbol{w}$ and the margin. Taking $b = 0$, find the slack $s_k = 1 - y_k(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_k - b)$ of each point.

$\mathcal{L}(\alpha) = 2\alpha - 4\alpha^2$ increases up to $\alpha = \frac{1}{4}$, which is now not allowed, so $\alpha = C = 0.1$. Then $\boldsymbol{w} = 0.1(1,1)^\mathsf{T} + 0.1(1,1)^\mathsf{T} = (0.2,0.2)^\mathsf{T}$ and the margin is $1/\|\boldsymbol{w}\| = 1/(0.2\sqrt{2}) \approx 3.54$, wider than before. Each point has $y_k \boldsymbol{w}^\mathsf{T}\boldsymbol{x}_k = 0.4$, so $s_k = 0.6$: both points sit inside the margin, though they are still classified correctly. A small C makes slack cheap, so the SVM buys a wider margin with it.
(For the curious: here b is not fixed. Any $|b| \le 0.6$ gives $s_1 + s_2 = 1.2$ and the same objective. Symmetry picks $b = 0$.)

[2 marks]

Answers

End of question 3

4 A function $K(\boldsymbol{x},\boldsymbol{y})$ is a valid (positive semi-definite) kernel if it can be written as $K(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{\phi}(\boldsymbol{x})^\mathsf{T}\boldsymbol{\phi}(\boldsymbol{y})$ for some real feature vector $\boldsymbol{\phi}(\boldsymbol{x})$. Equivalently, for any points $\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n$ the Gram matrix with elements $K_{ij} = K(\boldsymbol{x}_i,\boldsymbol{x}_j)$ is positive semi-definite.

(a) For $\boldsymbol{x},\boldsymbol{y} \in \mathbb{R}^2$, find a feature vector $\boldsymbol{\phi}(\boldsymbol{x})$ with six components such that $(1 + \boldsymbol{x}^\mathsf{T}\boldsymbol{y})^2 = \boldsymbol{\phi}(\boldsymbol{x})^\mathsf{T}\boldsymbol{\phi}(\boldsymbol{y})$.

Expanding,

$$(1 + x_1y_1 + x_2y_2)^2 = 1 + 2x_1y_1 + 2x_2y_2 + x_1^2y_1^2 + x_2^2y_2^2 + 2x_1x_2y_1y_2,$$

so $\boldsymbol{\phi}(\boldsymbol{x}) = (1,\ \sqrt{2}x_1,\ \sqrt{2}x_2,\ x_1^2,\ x_2^2,\ \sqrt{2}x_1x_2)^\mathsf{T}$. The features are all the monomials of degree up to two.

[2 marks]

(b) Write down the Gram matrix of the linear kernel $K(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{x}^\mathsf{T}\boldsymbol{y}$ for the three points $\boldsymbol{x}_1 = (1,0)^\mathsf{T}$, $\boldsymbol{x}_2 = (0,1)^\mathsf{T}$ and $\boldsymbol{x}_3 = (1,1)^\mathsf{T}$, and find its eigenvalues. Why is one of them zero?

$$\mathbf{K} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \qquad \det(\mathbf{K} - \lambda\mathbf{I}) = -\lambda(\lambda - 1)(\lambda - 3),$$

so the eigenvalues are 0, 1 and 3, all non-negative. The zero comes from $\boldsymbol{x}_3 = \boldsymbol{x}_1 + \boldsymbol{x}_2$: then $\mathbf{K}(1,1,-1)^\mathsf{T} = \mathbf{0}$. Three points in a 2-dimensional feature space must be linearly dependent.

[2 marks]

(c) Show that every valid kernel has $K(\boldsymbol{x},\boldsymbol{x}) \geq 0$. Hence show that $K(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{x}^\mathsf{T}\boldsymbol{y} - 1$ is not a valid kernel.

$K(\boldsymbol{x},\boldsymbol{x}) = \boldsymbol{\phi}(\boldsymbol{x})^\mathsf{T}\boldsymbol{\phi}(\boldsymbol{x}) = \|\boldsymbol{\phi}(\boldsymbol{x})\|^2 \geq 0$. But $\mathbf{0}^\mathsf{T}\mathbf{0} - 1 = -1 < 0$.

[2 marks]

(d) Show that $K(\boldsymbol{x},\boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|^2$ is not a valid kernel, even though $K(\boldsymbol{x},\boldsymbol{x}) \geq 0$. (Hint: look at the Gram matrix of two different points.)

For $\boldsymbol{x} \neq \boldsymbol{y}$ let $d = \|\boldsymbol{x} - \boldsymbol{y}\|^2 > 0$. The Gram matrix is $\begin{pmatrix} 0 & d \\ d & 0 \end{pmatrix}$, with eigenvalues $\pm d$. The eigenvalue $-d$ is negative, so the matrix is not positive semi-definite. (Equivalently, $\boldsymbol{v} = (1,-1)^\mathsf{T}$ gives $\boldsymbol{v}^\mathsf{T}\mathbf{K}\boldsymbol{v} = -2d < 0$.)

[2 marks]

Answers

(e) Using the rules from the lecture (sums, positive multiples, products and exponentials of valid kernels are valid), show that $K(\boldsymbol{x},\boldsymbol{y}) = 3(\boldsymbol{x}^\mathsf{T}\boldsymbol{y})^2 + \exp(\boldsymbol{x}^\mathsf{T}\boldsymbol{y})$ is a valid kernel.

$\boldsymbol{x}^\mathsf{T}\boldsymbol{y}$ is valid, with $\boldsymbol{\phi}(\boldsymbol{x}) = \boldsymbol{x}$. Its product with itself, $(\boldsymbol{x}^\mathsf{T}\boldsymbol{y})^2$, is valid, and so is 3 times it. The exponential of a valid kernel, $\exp(\boldsymbol{x}^\mathsf{T}\boldsymbol{y})$, is valid, and the sum of the two valid kernels is valid.

[2 marks]

Answers

End of question 4

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